Gautham Manuru Prabhu

gauthammprabhu2002@gmail.com | (+91) 7892184778 | Website | Portfolio | Google Scholar | GitHub | LinkedIn

Education

Manipal Institute of Technology, Manipal 2020 – 2024
B.Tech. in Computer Science and Engineering *CGPA: **8.91/10***

Work Experience

Cisco Systems Jan 2024 – Present
Software Engineering Intern → Software Engineer I → Software Engineer II *Bengaluru, India*

Software Engineer II, AI Acceleration Team *Oct 2025 – Present*

- Lead developer of an agentic test-triage platform (RAG, LLM diagnosis, MCP) deployed in factories worldwide; it finds root causes for **about 70% of previously undiagnosed** failures.
- Contributed to a support-queue agent that resolves **30% of routine requests**; presented it to Cisco's Chief Supply Chain Officer. Both systems cut case volume **35%** and MTTR **40%**, saving **$450,000** a year.
- Took the platform from prototype to production with security, platform, infrastructure, and transformation teams, and turned test and quality engineers' manual triage steps into agent workflows.
- Mentor **5+ junior engineers** on software engineering, agent design, code review, and operations.
- Received **25+ Cisco Connected Recognition awards** for technical contribution, mentoring, and volunteering.

Software Engineer I, Supply Chain Transformation *Aug 2024 – Sep 2025*

- Built Quality Transformation metrics engines tracking **10+ quality metrics** across **200+ product lines**.
- Added agentic AI to failure analysis on factory test lines and built a failure-classification suite that groups test failures by likely cause before engineers review them.
- Split legacy monoliths into **10+ FastAPI microservices** on Kubernetes across **8 global sites**.
- Deployed a computer-vision inspection system on assembly lines in multiple global factories that flags edge gaps in connectors from camera images, replacing manual visual checks.

Software Engineering Intern, Supply Chain Organization *Jan 2024 – Jun 2024*

- Rebuilt a **Python 2** workflow as microservices with CI/CD, cutting deployment from **6 hours to 4 minutes**.
- Built a supplier-name standardization pipeline that improved consistency across **50,000+ supplier records**.

Research Experience

MiCoSyS Lab, San José State University March 2026 – Present
Research Intern | Advisor: Dr. Saptarshi Sengupta *Remote*

- Tested whether LLM-generated semantics improve temporal link prediction on DTGB text-attributed graphs (Enron, ICEWS1819) once all models are evaluated under one common protocol.
- Reproduced CROSS, a reported state-of-the-art LLM-augmented model, and matched its published results; compared it with DyGFormer and a lightweight structure-aware gate (SAGE) under a **100-negative** protocol over **5 seeds**.
- Found the LLM gains depend on protocol and dataset: CROSS helped most on unseen (inductive) nodes but turned seed-dependent on ICEWS1819, SAGE matched it, and larger text encoders did not help.
- **First-author poster** accepted at the Stanford Graph Learning Workshop (SNAP) [W1]; A* conference manuscript in preparation.

Medical Informatics Lab, IIT Kharagpur Summer 2023
Research Intern | Advisor: Dr. Subhamoy Mandal *Remote*

- Built dual-encoder, cross-attention vision models to detect age-related macular degeneration from thermal and structural OCT, robust to missing modalities and inconsistent resolutions.
- Studied image fusion across RGB, HSV, and thermal representations to find which inputs best support AMD detection; presented the work at the Digital Health Symposium, IIT Kharagpur [T1].

Cybersecurity and Quantum Computing Group, MIT Manipal 2022 – 2023
Undergraduate Researcher | Advisor: Dr. Vivekananda Bhat K. *Manipal*

- Benchmarked QSVC, Pegasos QSVC, and quanvolutional networks for ECG-image cardiovascular classification; the latter reached **97.31% accuracy**, **3.88 percentage points** above a comparable CNN.
- Extended Pegasos QSVC, a quantum-kernel SVM with a Pegasos sub-gradient solver, from binary to four-class classification with six one-vs-one models; tuned feature maps, qubit counts, and circuit depth.
- Supported by a **$10,200 MeitY–AWS QCAL grant**; published in *IEEE Access* [1].

Biometrics and Software Engineering Group, MIT Manipal 2022 – 2023
Undergraduate Researcher | Advisor: Dr. Srikanth Prabhu *Manipal*

- Co-developed EyeEncrypt, secure retinal vessel segmentation for **3,000+ fundus images** (Diffie–Hellman, AES-256).
- Demonstrated it to Kasturba Medical College's ophthalmologists; published at ATIS 2022 [4] and in a review [5].

Department of Information and Communication Technology, MIT Manipal 2022 – 2023
Research Intern | Advisor: Dr. Nisha P. Shetty *Manipal*

- Fine-tuned and compared BERT and XLNet on the **15,073-tweet** COVID-19 ANTi-Vax dataset to separate misinformation from general vaccine discussion, and analyzed the precision-recall trade-offs between them.
- Linked Reddit authors by post-comment cosine similarity and used NetworkX to find misinformation clusters [2].

Publications

Six peer-reviewed publications and one accepted workshop poster; 60+ citations (Google Scholar).

[1] S. Prabhu, S. Gupta, **G. M. Prabhu**, A. V. Dhanuka, K. V. Bhat. "QuCardio: Application of quantum machine learning for detection of cardiovascular diseases." *IEEE Access*, vol. 11, pp. 136122–136135, 2023. [doi]

[2] C. Rao, **G. M. Prabhu**, A. R. Kumar, S. Gupta, N. P. Shetty. "Addressing vaccine misinformation on social media by leveraging transformers and user association dynamics." *Procedia Computer Science*, vol. 235, 2024. [doi]

[3] **G. M. Prabhu**, T. Gupta, M. V. Srujan, A. R. Soumya, A. Palorkar, A. Chowdhury. "VIKAS: A multimodal framework to aid in effective disaster management." *ATIS 2022*, Springer CCIS, vol. 1804, 2023. [doi]

[4] G. Hegde, S. Gupta, **G. M. Prabhu**, S. V. Bhandary. "EyeEncrypt: A cyber-secured framework for retinal image segmentation." *ATIS 2022*, Springer CCIS, vol. 1804, 2023. [doi]

[5] G. Hegde, S. Prabhu, S. Gupta, **G. M. Prabhu**, A. Palorkar, M. V. Srujan, S. V. Bhandary. "A systematic review of deep learning approaches for vessel segmentation in retinal fundus images." *Journal of Physics: Conference Series*, 2023. [doi]

[6] A. A. Iyer, **G. M. Prabhu**, T. Gupta, S. Deshmukh, R. Rivankar. "Exploring design optimisation techniques of a radio telescope using fixed costing constraints." *Journal of Physics: Conference Series*, 2023. [doi]

[W1] **G. M. Prabhu**, S. Gupta, M. Choudhary, A. Nagarajan, K. Potika, S. Sengupta. "On link prediction using node-pair LLM semantics and structure in temporal text-attributed graphs." *Stanford Graph Learning Workshop (SNAP)*, 2026 (**first-author** poster).

Talks, Summer Schools, and Workshops

[T1] **G. M. Prabhu**, S. Gupta, S. Rathinam. "Enhanced analytics and procedural decision-making for ocular diagnostics using multi-modal data." *Digital Health Symposium*, IIT Kharagpur, 2023.

[T2] **G. M. Prabhu**, S. Gupta. "SatelTensor: Satellite data exploration via tensor decomposition." *TCML*, IISc, 2023.

[S1] ACM Winter School on Optimization for Machine Learning and Operations Research, IIT Goa, 2022.

[S2] Summer School on Dynamic Resource Allocation, Centre for Networked Intelligence, IISc Bengaluru, 2023.

Honors and Achievements

- **Winner**, Offensive Summit Capture the Flag (CTF) security competition, Cisco 2026
- **25+** Cisco Connected Recognition awards 2024 – 2026
- **Second runner-up**, Cisco Intern Case Study Competition 2024
- Reviewer, International Conference on Machine Learning and Data Engineering (ICMLDE) 2025
- Undergraduate Research Sponsorship Grant, Manipal Academy of Higher Education 2023
- **$10,200** MeitY–AWS Quantum Computing Applications Lab (QCAL) research grant for QuCardio [1] 2023
- Grand finalist, Global Quantum Science and Technology Hackathon (top **16** of **1,626** teams) 2022
- Grand finalist and team lead, Smart India Hackathon (**160,000+** students, **2,235+** institutes) [3] 2022
- NTSE State Scholar, Government of Karnataka (rank **21** among **151,000+** candidates) 2018

Leadership

- Lead Volunteer, Cisco Emerging Talent Recruitment, University Connect Track 2026 – Present
- Track Lead Volunteer, Cisco Global Supply Chain, Innovation Track 2026 – Present
- Co-founder and Technical Head, Project Kalpana 2022 – 2023
- General Secretary and Treasurer, ACM Manipal Student Chapter 2022 – 2023
- Technical Head, The Astronomy Club of Manipal 2021 – 2023

Skills and Interests

Research	Graph ML, temporal GNNs, LLMs, PEFT (LoRA), RAG, agentic systems, quantum ML
Languages & ML	Python, C++, SQL, Bash, PyTorch, PyG, DGL, Hugging Face, Qiskit, OpenCV, NetworkX
LLM & Agents	LangChain, LangGraph, LangSmith, MCP, Ragas, DeepEval, Pinecone, Weaviate
Backend & Cloud	FastAPI, microservices, Docker, Kubernetes, AWS, GCP, Jenkins, GitHub Actions, Git
Databases	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Neo4j
Interests	Volunteering, scuba diving, quizzing, trekking, astronomy